

Instructions

This mock mirrors the Infosys SP/DSE online assessment structure. Attempt it in one timed sitting; review using the highlighted answers.

Section 1 - Logical Reasoning | Section 2 - Quantitative Ability | Section 3 - Verbal Ability | Section 4 - Pseudocode/Technical | Section 5 - Coding (2 problems).

This is Mock Test 3 of 3; each mock uses a different question set.

Section 1 - Logical Reasoning

Q1. All A are B. No B is C. Conclusion: No A is C.

(A) Follows ✓

(B) Does not follow

(C) Cannot be determined

(D) None

Answer: (A) Follows

Authenticity: Frequently Repeated

Q2. All roses are flowers. Some flowers fade quickly. Conclusion: Some roses fade quickly.

(A) Follows

(B) Does not follow ✓

(C) Always true

(D) None

Answer: (B) Does not follow

Authenticity: Frequently Repeated

Q3. Next term: 1, 3, 6, 10, 15, ?

(A) 18

(B) 20

(C) 21 ✓

(D) 24

Answer: (C) 21

Authenticity: Frequently Repeated

Q4. Next term: 2, 3, 5, 7, 11, ?

(A) 12

(B) 13 ✓

(C) 14

(D) 15

Answer: (B) 13

Authenticity: Frequently Repeated

Q5. If CAT is DBU (each +1), then DOG is:

(A) EPH ✓

(B) EPG

(C) FPH

(D) DPH

Answer: (A) EPH

Authenticity: Community Reported

Q6. If A=1,B=2,..., value of BAD is:

(A) 6

(B) 7 ✓

(C) 8

(D) 5

Answer: (B) 7

Authenticity: Community Reported

Q7. 'She is the daughter of my grandfather's only son.' She is the speaker's:

(A) Mother

(B) Sister ✓

(C) Aunt

(D) Cousin

Answer: (B) Sister

Authenticity: Community Reported

Q8. Facing north, turn 135 deg clockwise then 180 deg anticlockwise. Now facing:

(A) North-East

(B) South-East

(C) North-West ✓

(D) South-West

Answer: (C) North-West

Authenticity: Research-Backed Company Style

Q9. 8 : 64 :: 9 : ?

(A) 72

(B) 81 ✓

(C) 729

(D) 90

Answer: (B) 81

Authenticity: Community Reported

Q10. Odd one out: 8, 27, 64, 100.

(A) 8

(B) 27

(C) 64

(D) 100 ✓

Answer: (D) 100

Authenticity: Community Reported

Q11. If today is Monday, the day after 61 days is:

(A) Friday

(B) Saturday ✓

(C) Sunday

(D) Thursday

Answer: (B) Saturday

Authenticity: Community Reported

Q12. Ranked 8th from top and 22nd from bottom. Total students:

(A) 28

(B) 29 ✓

(C) 30

(D) 31

Answer: (B) 29

Authenticity: Community Reported

Section 2 - Quantitative Ability

Q1. Find the next term: 3, 6, 12, 24, ?

(A) 36

(B) 48 ✓

(C) 30

(D) 42

Answer: (B) 48

Authenticity: Frequently Repeated

Q2. Find the next term: 100, 96, 88, 76, 60, ?

(A) 40 ✓

(B) 48

(C) 44

(D) 50

Answer: (A) 40

Authenticity: Frequently Repeated

Q3. 30% of a number is 135. Find the number.

(A) 405

(B) 450 ✓

(C) 500

(D) 350

Answer: (B) 450

Authenticity: Frequently Repeated

Q4. A number is increased by 20% then decreased by 20%. Net change:

- (A) No change
- (B) 4% increase

(C) 4% decrease ✓

- (D) 2% decrease

Answer: (C) 4% decrease

Authenticity: Frequently Repeated

Q5. Marked price is 25% above CP; a 10% discount is given. Profit percent:

- (A) 15%

(B) 12.5% ✓

- (C) 10%
- (D) 13.5%

Answer: (B) 12.5%

Authenticity: Community Reported

Q6. Two articles sold at Rs. 990 each, one at 10% profit and one at 10% loss. Net result:

- (A) No profit no loss

(B) Loss of Rs. 20 ✓

- (C) Profit of Rs. 20
- (D) Loss of Rs. 10

Answer: (B) Loss of Rs. 20

Authenticity: Community Reported

Q7. 10 men finish in 8 days. 8 men take:

(A) 10 days ✓

- (B) 6 days
- (C) 12 days
- (D) 9 days

Answer: (A) 10 days

Authenticity: Community Reported

Q8. Tap A fills in 4 h, tap B empties in 9 h. Both open, time to fill:

- (A) 6 h

(B) 7.2 h ✓

- (C) 8 h
- (D) 5 h

Answer: (B) 7.2 h

Authenticity: Community Reported

Q9. Distance covered at 40 km/h one way, 60 km/h back. Average speed:

- (A) 50 km/h

(B) 48 km/h ✓

- (C) 45 km/h
- (D) 52 km/h

Answer: (B) 48 km/h

Authenticity: Community Reported

Q10. Walking at $\frac{5}{6}$ of usual speed a man is 10 min late. Usual time:

(A) 40 min

(B) 50 min ✓

(C) 60 min

(D) 45 min

Answer: (B) 50 min

Authenticity: Research-Backed Company Style

Q11. Mean proportional between 4 and 16:

(A) 6

(B) 8 ✓

(C) 10

(D) 12

Answer: (B) 8

Authenticity: Community Reported

Q12. Average of first 10 natural numbers:

(A) 5

(B) 5.5 ✓

(C) 6

(D) 4.5

Answer: (B) 5.5

Authenticity: Community Reported

Q13. A sum doubles in 8 years at simple interest. Rate:

(A) 10%

(B) 12.5% ✓

(C) 15%

(D) 8%

Answer: (B) 12.5%

Authenticity: Community Reported

Q14. A sum amounts to Rs. 1210 in 2 years and Rs. 1331 in 3 years (CI). Rate:

(A) 8%

(B) 10% ✓

(C) 12%

(D) 11%

Answer: (B) 10%

Authenticity: Research-Backed Company Style

Section 3 - Verbal Ability

Q1. Synonym of DILIGENT:

- (A) Lazy
- (B) Careless

(C) Hardworking ✓

- (D) Slow

Answer: (C) Hardworking

Authenticity: Community Reported

Q2. Synonym of LUCID:

- (A) Confusing

(B) Clear ✓

- (C) Dark
- (D) Heavy

Answer: (B) Clear

Authenticity: Research-Backed Company Style

Q3. Antonym of EXPAND:

- (A) Grow
- (B) Stretch

(C) Contract ✓

- (D) Widen

Answer: (C) Contract

Authenticity: Community Reported

Q4. Antonym of GENEROUS:

- (A) Kind

(B) Stingy ✓

- (C) Warm
- (D) Noble

Answer: (B) Stingy

Authenticity: Research-Backed Company Style

Q5. Choose the correct sentence (comparative preposition):

- (A) She is senior than me.

(B) She is senior to me. ✓

- (C) She is more senior than me.
- (D) She is senior from me.

Answer: (B) She is senior to me.

Authenticity: Community Reported

Q6. The book is ____ the table.

- (A) in
- (B) at

(C) on ✓

- (D) into

Answer: (C) on

Authenticity: Community Reported

Q7. (Same passage) What is described as essential?

(A) More fossil fuels

(B) Storage technology ✓

(C) Higher prices

(D) Government bans

Answer: (B) Storage technology

Authenticity: Research-Backed Company Style

Section 4 - Pseudocode / Technical

Q1. Which data structure follows LIFO?

(A) Queue

(B) Stack ✓

(C) Linked List

(D) Heap

Answer: (B) Stack

Authenticity: Community Reported

Q2. Average-case lookup time in a hash table:

(A) $O(1)$ ✓

(B) $O(n)$

(C) $O(\log n)$

(D) $O(n \log n)$

Answer: (A) $O(1)$

Authenticity: Community Reported

Q3. The recurrence $T(n) = 2T(n/2) + n$ solves to:

(A) $O(n)$

(B) $O(n \log n)$ ✓

(C) $O(n^2)$

(D) $O(\log n)$

Answer: (B) $O(n \log n)$

Authenticity: Frequently Repeated

Q4. $x=10$; $y=20$; $x=x+y$; $y=x-y$; $x=x-y$; Final x , y :

(A) $x=10, y=20$

(B) $x=20, y=10$ ✓

(C) $x=30, y=10$

(D) $x=20, y=20$

Answer: (B) $x=20, y=10$

Authenticity: Frequently Repeated

Q5. count=0; n=18; while(n>0){ count++; n=n/2; } (integer div). count:

(A) 4

(B) 5 ✓

(C) 6

(D) 3

Answer: (B) 5

Authenticity: Frequently Repeated

Section 5 - Coding

Problem 1: Move All Zeroes to End

Difficulty: *Easy* | Pattern: *In-place Two Pointers* | Authenticity: *Community Reported*

Problem Statement

Move all zeroes to the end while keeping the order of non-zero elements.

Input Format

Line 1: N. Line 2: N integers.

Output Format

Modified array, space-separated.

Constraints

$1 \leq N \leq 10^5$

Sample Input

```
7
0 1 0 3 12 0 5
```

Sample Output

```
1 3 12 5 0 0 0
```

Python Solution

```
def move_zeroes(arr):
    pos = 0
    for x in arr:
        if x != 0:
            arr[pos] = x; pos += 1
    while pos < len(arr):
        arr[pos] = 0; pos += 1
    return arr

print(' '.join(map(str, move_zeroes([0, 1, 0, 3, 12, 0, 5]))))
```

Step-by-Step Explanation

A write pointer copies non-zero values forward; remaining slots are filled with zeroes.

Dry Run

Non-zeros 1,3,12,5 written first; then zeroes fill positions 4-6.

Time Complexity: **O(N)** | Space Complexity: **O(1)**

Problem 2: Count Primes (Sieve of Eratosthenes)

Difficulty: **Medium** | Pattern: **Sieve** | Authenticity: **Community Reported**

Problem Statement

Count the number of prime numbers strictly less than n .

Input Format

A single integer n .

Output Format

Count of primes below n .

Constraints

$0 \leq n \leq 5 \cdot 10^6$

Sample Input

10

Sample Output

4

Python Solution

```
def count_primes(n):
    if n < 2:
        return 0
    sieve = [True] * n
    sieve[0] = sieve[1] = False
    for i in range(2, int(n ** 0.5) + 1):
        if sieve[i]:
            for j in range(i * i, n, i):
                sieve[j] = False
    return sum(sieve)

print(count_primes(10))
```

Step-by-Step Explanation

Mark multiples of each prime as composite; the survivors are primes. Below 10: 2,3,5,7.

Dry Run

4 primes remain.

Time Complexity: **O(n log log n)** | Space Complexity: **O(n)**

Full-Length Mock Test 3

Applied Role(s):

Digital Specialist Engineer (Trainee) | Specialist Programmer L1 / L2 / L3 (Trainee)

Online Assessment Preparation Material

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Authenticity Legend & How to Read This Document

Every question and problem in this document is labelled with an **authenticity classification** so you know how closely it maps to what candidates have actually reported. Labels are assigned conservatively based on research; authored practice questions are never presented as real exam questions.

Label	Meaning
Verified Previous-Year	The specific problem/pattern is documented from actual reported assessment experiences.
Frequently Repeated	The topic or pattern type is documented as recurring across multiple hiring cycles.
Community Reported	A widely shared, role-relevant pattern reported across placement communities and forums.
Research-Backed Company Style	Authored to match the researched difficulty, style and pattern - practice, not a real past question.

The correct option in every MCQ is highlighted in green and also restated below the question. Study the pattern, not just the answer.